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Roll Number: 48
Section: BS CS A

LAB # 02

TASK 1 — PROGRAM 1

hello.cpp | C++

```cpp
#include <iostream>
using namespace std;
int main()
{
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Output | Debug Console

```
A
B
C  (sample input; program waits for one character)
```

TASK 1 — PROGRAM 2

hello.cpp | C++

```cpp
#include <iostream>
using namespace std;
int main()
{
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    return 0;
}
```

Output | Debug Console

```
A
B
```

TASK 1 — PROGRAM 3

hello.cpp | C++

```cpp
#include <iostream>
using namespace std;
int main()
{
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    return 0;
}
```

Output | Debug Console

```
65
66
```

TASK 2 — ARITHMETIC OPERATIONS

hello.cpp | C++

```cpp
#include <iostream>
using namespace std;
int main()
{
    int a, b;
    cout << "Enter two integers: ";
    cin >> a >> b;
    cout << "Addition = " << a + b << endl;
    cout << "Subtraction = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    cout << "Division = " << a / b << endl;
    return 0;
}
```

Output | Debug Console

```
Enter two integers: 12 4
Addition = 16
Subtraction = 8
Multiplication = 48
Division = 3
```

TASK 3 — GIVEN PROGRAM OUTPUT

hello.cpp | C++

```cpp
#include <iostream>
#include <cstdlib>
using namespace std;
int main()
{
    int val1, val2, val3;
    cin >> val1 >> val2 >> val3;
    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
    system("PAUSE");
    return 0;
}
```

Output | Debug Console

```
Input: 10 20 30
10
20
30
30
20
10
Press any key to continue . . .
```

TASK 4 — DISTANCE BETWEEN TWO POINTS

hello.cpp | C++

```cpp
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
    double x1, y1, x2, y2, d;
    cout << "Enter x1 y1 x2 y2: ";
    cin >> x1 >> y1 >> x2 >> y2;
    d = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
    cout << "Distance = " << d << endl;
    return 0;
}
```

Output | Debug Console

```
Enter x1 y1 x2 y2: 3 6 5 8
Distance = 2.82843
```

TASK 5 — AREA OF A CIRCLE

hello.cpp | C++

```cpp
#include <iostream>
using namespace std;
int main()
{
    float pi = 3.14159;
    float radius = 5.0;
    float area = pi * radius * radius;
    cout << "The area of a circle with radius " << radius
         << " is " << area << endl;
    return 0;
}
```

Output | Debug Console

```
The area of a circle with radius 5 is 78.5397
```

TASK 5 — AREA OF A RECTANGLE

hello.cpp | C++

```cpp
#include <iostream>
using namespace std;
int main()
{
    double length, width;
    cin >> length >> width;
    cout << "Area = " << length * width << endl;
    return 0;
}
```

Output | Debug Console

```
Input: 5 3
Area = 15
```

TASK 5 — AREA OF A TRIANGLE

hello.cpp | C++

```cpp
#include <iostream>
using namespace std;
int main()
{
    double base, height;
    cin >> base >> height;
    cout << "Area = " << base * height * 0.5 << endl;
    return 0;
}
```

Output | Debug Console

```
Input: 8 4
Area = 16
```

TASK 6 — ITEM BILL, TAX AND TOTAL

hello.cpp | C++

```cpp
#include <iostream>
#include <iomanip>
using namespace std;
int main()
{
    double p1 = 12.95, p2 = 24.95, p3 = 6.95;
    double p4 = 14.95, p5 = 3.95;
    double subtotal = p1 + p2 + p3 + p4 + p5;
    double tax = subtotal * 0.06;
    double total = subtotal + tax;
    cout << fixed << setprecision(2);
    cout << "Item 1: $" << p1 << endl;
    cout << "Item 2: $" << p2 << endl;
    cout << "Item 3: $" << p3 << endl;
    cout << "Item 4: $" << p4 << endl;
    cout << "Item 5: $" << p5 << endl;
    cout << "Subtotal: $" << subtotal << endl;
    cout << "Sales tax (6%): $" << tax << endl;
    cout << "Total: $" << total << endl;
    return 0;
}
```

Output | Debug Console

```
Item 1: $12.95
Item 2: $24.95
Item 3: $6.95
Item 4: $14.95
Item 5: $3.95
Subtotal: $63.75
Sales tax (6%): $3.83
Total: $67.58
```

BONUS — SEPARATE FIVE DIGITS

hello.cpp | C++

```cpp
#include <iostream>
using namespace std;
int main()
{
    int n;
    cout << "Enter a five-digit integer: ";
    cin >> n;
    cout << n / 10000 << "   ";
    cout << (n / 1000) % 10 << "   ";
    cout << (n / 100) % 10 << "   ";
    cout << (n / 10) % 10 << "   ";
    cout << n % 10 << endl;
    return 0;
}
```

Output | Debug Console

```
Enter a five-digit integer: 12345
1   2   3   4   5
```

HOME TASK 1 — SWAP TWO VARIABLES

hello.cpp | C++

```cpp
#include <iostream>
using namespace std;
int main()
{
    int a, b, temp;
    cout << "Enter two values: ";
    cin >> a >> b;
    temp = a;
    a = b;
    b = temp;
    cout << "After swapping: " << a << " " << b << endl;
    return 0;
}
```

Output | Debug Console

```
Enter two values: 5 9
After swapping: 9 5
```

HOME TASK 2 — LOWERCASE TO UPPERCASE

hello.cpp | C++

```cpp
#include <iostream>
using namespace std;
int main()
{
    char ch;
    cout << "enter small alphabet: ";
    cin >> ch;
    ch = ch - 32;
    cout << "uppercase alphabet " << ch << endl;
    return 0;
}
```

Output | Debug Console

```
enter small alphabet: g
uppercase alphabet G
```